

Table 1 Providers of immunisations to 204 inpatients according to where patients lived

	Total No	No (%) of inpatients immunised by various providers				
		Private doctor only	Local council only	Department of Health only	More than one	None
Sydney	160	108 (68)	11 (7)	15 (9)	25 (16)	1 (0.6)
Elsewhere	44	28 (65)	1 (2)	9 (20)	4 (9)	2 (5)
Total	204 (100)	136 (67)	12 (6)	22 (11)	29 (14)	3 (1)

viewed and immunisation data verified for all of them. There were 87 (43%) girls and 117 boys, mean age 11.7 months. Of the 204 children, 159 (78%) had received all the required first year doses of triple antigen, and of the 33 children old enough to have received the 18 month booster dose, 15 had received this dose as triple antigen, and three as combined diphtheria-tetanus. Thus 144 (71%) were appropriately immunised for age with triple antigen, and an additional 20 (9%) had received some doses of combined diphtheria-tetanus instead of triple antigen. One hundred and seventy six (86%) children had received the correct number of oral polio vaccine doses, and 48 (81%) of the 59 children aged 16 months or older had received measles-mumps/measles-mumps-rubella. Overall, only 135 of the 204 children (66%) had complied fully with the recommended schedule.

COMPARISON OF INPATIENTS WITH OUTPATIENTS

One hundred forty four of 204 inpatients (71%) had received all triple antigen doses for age compared with 93 of 111 outpatients (84%). This difference is significant ($\chi^2=6.0$, $p<0.02$). In addition, 176 inpatients (86%) had received all oral polio vaccine doses for age, compared with 106 outpatients (95%). This difference is significant ($\chi^2=5.6$, $p<0.02$). The compliance with measles-mumps immunisation in the two groups was similar ($\chi^2=0.0$, $p>0.95$).

USE OF THE PERSONAL HEALTH RECORD

When 24 of the 204 children either born outside New South Wales and the Australian Capital Territory, or before the introduction of the personal health record, were excluded from the analysis, 176 (98%) of a possible 180 children had a personal health record; in the remainder the mothers had not been given them. Many parents, however, did not bring the personal health record into hospital at the time of admission, so that the personal health record was available for only 46% of children. Of 88 children admitted urgently (the casualty department, doctor's rooms, or interhospital transfer) only 30 (34%) had their personal health record on admission, compared with 51 of 87 children (59%) admitted electively ($\chi^2=9.6$, $p<0.01$).

PROVIDERS OF IMMUNISATIONS

For analysis of usage patterns, the 204 children were divided into two groups according to where they lived: 'Sydney' (that is, those residing in central, eastern, northern, southern, western or south western health areas) and

Table 2 Accuracy of documentation of inpatients' immunisation state in medical records when compared with immunisation records held by parents or providers of immunisations

Accuracy of hospital medical record	No of patients with complete immunisations verified	No of patients with incomplete immunisations verified
Correct:detailed	58	26
Correct:abbreviated	65	6
Incorrect	5	14
Undocumented	27	3
Total	155	49

'elsewhere'. Immunisation providers were grouped for purposes of analysis into: 'private doctors' (general practitioners, medical practices, and paediatricians), 'local council', and 'Department of Health facilities' (comprising hospital immunisation clinics and early childhood centres) (table 1). There were 160 children living in 'Sydney' and 44 living 'elsewhere'. Most used private doctors exclusively for their immunisations, and a relatively small number used only local council clinics. When the users of multiple services were taken into account, private doctors were used for some or all immunisations by 131 Sydney residents (82%), and by 31 of those living elsewhere (70%). Similarly, local councils were used for some or all immunisations by 30 Sydney residents (19%) and by four of those living elsewhere (9%).

ACCURACY OF MEDICAL NOTES

The adequacy of documentation of immunisations in the children's medical notes compared with other records is shown in table 2. Overall, documentation of immunisations was correct in 155 medical records (76%), and incorrect or absent in 49 (24%). Immunisation state was incorrect or undocumented in 32 (19%) of the 155 medical notes of the children who were appropriately immunised and in 17 of 49 (35%) of those incompletely immunised.

Discussion

These data show that the population of young children attending a children's hospital had a level of immunisation uptake lower than the 90-95% required for herd immunity to prevail in the general child population.^{16 17} The casualty and outpatients survey showed vaccine uptakes of 79% for measles-mumps, 90% for first year triple antigen, and 95% for oral polio vaccine. These data are difficult to compare with other studies because of differences in the populations surveyed and in other methods used, but a similar survey of children in casualty